

A simultaneous quenched hard-edge limit for weighted hyperelliptic prime races

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Abstract

We study a weighted inert-versus-split prime race in the one-parameter hyperelliptic pencil $y^2 = \prod_{a=1}^{2g} (x - a)(x - t)$ over finite fields. For each curve, the normalized race sequence has a Cesàro endpoint law. In the critical scaling $m + 1/2 = \lambda / ((2g + 1) \log Q)$, we prove that these quenched laws, viewed as random elements of $\mathcal{P}_2(\mathbb{R})$, converge to the random law generated by a marked symplectic hard-edge process. The limit is simultaneous in genus and field size for odd square prime powers satisfying $\log Q / (g^2 \log(g + 2)) \rightarrow \infty$, and we give a completely specified sequence of fields. The limiting random measure and its positive-half-line mass are both nonconstant. The proof combines density-one linear independence, genus-explicit Frobenius equidistribution, heat-kernel smoothing, and direct W_2 control of the marked series. This is a large-field/large-genus bridge theorem, not a fixed-field microscopic result.

1 Introduction and statement of the theorem

Prime races over a function field turn the zeros of an arithmetic L -function into a finite collection of Frobenius angles. For a fixed curve, linear independence of those angles converts the endpoint variable into a random-phase cosine sum; for a family of curves, monodromy and equidistribution make the coefficients themselves random. These two kinds of randomness interact most sharply at the hard edge, where the weight is tuned so that the lowest Frobenius angles contribute at order one.

This paper isolates that interaction in one explicit family and one simultaneous limit. Fix $\lambda > 0$. For each $g \geq 2$, let Q_g be an odd square prime power of characteristic $p_g > 2g + 1$, put

$$f_g(x) = \prod_{a=1}^{2g} (x - a), \quad U_g = \mathbb{A}^1 \setminus \{1, \dots, 2g\},$$

and consider the pencil

$$C_{g,t} : y^2 = f_g(x)(x - t), \quad t \in U_g(\mathbb{F}_{Q_g}). \quad (1.1)$$

The curve has genus g . Its quadratic character $\chi_{g,t}$ is $+1$ at split prime polynomials and -1 at inert ones. Write

$$d_g = 2g + 1, \quad M_g = m_g + \frac{1}{2} = \frac{\lambda}{d_g \log Q_g}. \quad (1.2)$$

For an endpoint N , the normalized inert-minus-split race is

$$\mathcal{E}_{g,t,\lambda}(N) = 2(1 - Q_g^{-M_g})Q_g^{-M_g N} \left(- \sum_{\substack{P \text{ monic irreducible} \\ \deg P \leq N}} (\deg P) \chi_{g,t}(P) Q_g^{m_g \deg P} \right). \quad (1.3)$$

Write $\mathcal{P}_2(\mathbb{R})$ for the space of Borel probability measures on $\mathbb{R}$ with finite second moment, equipped with the quadratic Wasserstein metric W_2 .

For every parameter t , the empirical endpoint distribution of $\mathcal{E}_{g,t,\lambda}(N)$ exists; this follows from Kronecker–Weyl on the closure of the cyclic angle–parity orbit. We denote the genuine endpoint law by $\nu_{g,t,\lambda} \in \mathcal{P}_2(\mathbb{R})$. When the positive Frobenius angles together with π are linearly independent over $\mathbb{Q}$, that orbit is the full phase torus with an independent fair parity coordinate, giving the explicit law in (2.12).

The limiting object is a random probability measure. Let Π_{Sp} be the determinantal process on $[0, \infty)$ with kernel

$$K_{\text{Sp}}(x, y) = \frac{\sin \pi(x - y)}{\pi(x - y)} - \frac{\sin \pi(x + y)}{\pi(x + y)}. \quad (1.4)$$

For a locally finite configuration ξ , define

$$a_\lambda(y) = \frac{4\lambda}{\sqrt{\lambda^2 + (2\pi y)^2}}, \quad \nu_\lambda(\xi) = \text{Law}_\Phi \left(1 + \sum_{y \in \xi} a_\lambda(y) \cos \Phi_y \right), \quad (1.5)$$

where, conditionally on ξ , the phases are independent and uniform on $[0, 2\pi]$. The series converges in conditional L^2 almost surely. There are now two probability layers. The phases are integrated out to form the conditional, or quenched, measure; the uniform curve parameter t , and in the limit the point configuration Π_{Sp} , supply the outer randomness that makes this measure itself random. Thus $\implies$ below means weak convergence of the outer laws on the Polish space $(\mathcal{P}_2(\mathbb{R}), W_2)$. The coefficient sequences are naturally square-summable, and the shared-phase coupling controls their second moments exactly, which is why W_2 is the topology carried by the proof.

Theorem 1.1 (Simultaneous quenched critical law). *For every fixed $\lambda > 0$, suppose*

$$\frac{\log Q_g}{g^2 \log(g + 2)} \longrightarrow \infty. \quad (1.6)$$

Choose t_g uniformly from $U_g(\mathbb{F}_{Q_g})$. Then, as random elements of $(\mathcal{P}_2(\mathbb{R}), W_2)$,

$$\nu_{g,t_g,\lambda} \implies \nu_\lambda(\Pi_{\text{Sp}}). \quad (1.7)$$

The limiting random probability measure is not almost surely constant and is almost surely absolutely continuous.

Corollary 1.2 (Prime-race densities). *Under the hypotheses of theorem 1.1, the positive-half-line masses satisfy*

$$\nu_{g,t_g,\lambda}((0, \infty)) \implies \nu_\lambda(\Pi_{\text{Sp}})((0, \infty)), \quad (1.8)$$

and

$$\lim_{g \rightarrow \infty} \mathbb{E}_{t_g} \nu_{g,t_g,\lambda}((0, \infty)) = \mathbb{E} \nu_\lambda(\Pi_{\text{Sp}})((0, \infty)).$$

The limiting scalar $\nu_\lambda(\Pi_{\text{Sp}})((0, \infty))$ is not almost surely constant. With probability $1 - o(1)$, the left side of (1.8) is the natural density of positive endpoints in the original race.

Example 1.3 (An explicit admissible sequence). *The conclusions of theorem 1.1 and corollary 1.2 hold when p_g is the least prime greater than $2g + 1$ and*

$$Q_g = p_g^{2n_g}, \quad n_g = \min \{n \geq 1 : p_g^{2n} \geq \exp(g^2 \log^2(g + 2))\}. \quad (1.9)$$

Indeed, equation (3.8) verifies equation (1.6).

The order of limits in theorem 1.1 is not iterated: the genus, the characteristic, and the extension degree vary along one specified sequence. The square-field hypothesis is structural. It makes $\sqrt{Q_g}$ rational, so that a maximal Weyl-group Galois condition on the Frobenius polynomial implies the angle-plus- π independence required by the endpoint race.

The result is nevertheless not a fixed-field microscopic theorem. Condition (1.6) forces the field to grow very rapidly and lets quantitative Frobenius equidistribution dominate all genus-dependent smoothing costs. The fixed- Q , large-genus hard-edge problem remains outside the argument.

Relation to earlier work. Cha developed limiting distributions and Chebyshev bias for function-field prime races [4], with subsequent refinements and related family results in [7, 6, 2]. A particularly close direct race precedent is Bailleul–Devin–Keliher–Li [1]: they study the same hyperelliptic split-versus-inert setting, fixed-curve endpoint laws, parity centers, and the rarity of exceptional biases, but not a moving critical weight or a random measure-valued hard-edge limit. Recent work on ties and on high-trace hyperelliptic statistics further shows that neither function-field race phenomena nor simultaneous large-field/large-genus information is new by itself [3, 16].

Earlier race work therefore supplies fixed-curve scalar laws and bias statistics, while hyperelliptic-family work supplies Frobenius equidistribution and low-zero statistics. The object studied here retains the Frobenius configuration as outer randomness and proves convergence of the conditional law itself in W_2 , at the moving critical weight for which the lowest angles contribute at order one. The three bridges established are the exact weighted-race reduction, a genus-explicit transfer for a nonsmooth measure-valued test, and closed-edge ℓ^2 stability of the marked law.

The combination of generic Frobenius linear independence with family averaging also appears in Cha’s hyperelliptic Möbius study and in Perret–Gentil’s trace-function families [5, 17]; their statistics and limiting objects differ from the quenched critical law here. The hyperelliptic monodromy and quantitative equidistribution inputs used here are due to Katz–Sarnak [12]; density-one maximal Galois group and the relation-space criterion are due to Kowalski [14, 13, 15]. Low-zero statistics for hyperelliptic families have been studied in several regimes, including [10, 9]. Our contribution is the critical weighted-race calculation and the quantitative passage from the actual endpoint law to a nondegenerate random $\mathcal{P}_2(\mathbb{R})$ -valued hard-edge law in one simultaneous arithmetic regime. We do not claim that simultaneous large field and large genus, by themselves, are new.

Proof structure. The argument follows one chain:

$$\begin{aligned} \text{endpoint laws} &\xrightarrow{\text{LI}} \text{Frobenius-phase laws} \\ &\xrightarrow{\text{heat bridge}} \text{Haar USp}(2g) \text{ laws} \\ &\xrightarrow{\text{marked } W_2 \text{ stability}} \text{hard-edge quenched law.} \end{aligned}$$

Section 2 derives the exact random-phase law, including the two prime-square parity centers. Section 3 proves that the required linear independence fails on a set of density $o(1)$. Section 4 transfers the probability-measure-valued Frobenius functional to Haar $\text{USp}(2g)$, with every genus dependence visible. Section 5 proves the closed-hard-edge marked-law limit and both nondegeneracy assertions. Section 6 assembles the estimates. The two analytic tools most sensitive to normalization are proved separately in the appendices.

2 From the prime race to a quenched cosine law

In this section g, Q, t are fixed, and we suppress them from the notation. Put

$$d = 2g + 1, \quad M = m + \frac{1}{2} = \frac{\lambda}{d \log Q}, \quad r = Q^M = e^{\lambda/d} > 1. \quad (2.1)$$

The endpoint N tends to infinity before either g or Q varies. This order is important: all the error estimates below may depend on the fixed curve, but the limiting law that they identify is exact.

Write $D_{g,t}(x) = f_g(x)(x - t)$. The associated quadratic character is understood to vanish on polynomials not coprime to $D_{g,t}$. Its L -polynomial has the factorization

$$L(u, \chi_{g,t}) = \prod_{j=1}^{2g} (1 - \alpha_j u), \quad \alpha_j = Q^{1/2} e^{i\vartheta_j}, \quad \alpha_{g+j} = Q^{1/2} e^{-i\vartheta_j} \quad (2.2)$$

after a suitable ordering, where $0 \leq \vartheta_j \leq \pi$. For a monic polynomial F , write $\Lambda(F) = \deg P$ if $F = P^a$ for a monic irreducible P , and $\Lambda(F) = 0$ otherwise. Logarithmic differentiation of the Euler product and of (2.2) gives, coefficient by coefficient,

$$\Psi(k) := \sum_{\deg F=k} \Lambda(F) \chi_{g,t}(F) = - \sum_{j=1}^{2g} \alpha_j^k. \quad (2.3)$$

There are no omitted local terms here: at a prime dividing $D_{g,t}$ the quadratic Euler symbol is zero, and the corresponding Euler factor is one.

Define the unnormalised prime sum

$$A(N) := \sum_{\substack{P \text{ monic irreducible} \\ \deg P \leq N}} (\deg P) \chi_{g,t}(P) Q^{m \deg P}.$$

Separating the first powers in the von Mangoldt sum yields the exact identity

$$\begin{aligned}
-A(N) &= \sum_{k \leq N} \sum_{j=1}^{2g} (\alpha_j Q^m)^k + R_2(N) + R_{\geq 3}(N), \\
R_2(N) &:= \sum_{\substack{P \\ 2 \deg P \leq N}} (\deg P) \chi_{g,t}(P)^2 Q^{2m \deg P}, \\
R_{\geq 3}(N) &:= \sum_{\substack{P, a \geq 3 \\ a \deg P \leq N}} (\deg P) \chi_{g,t}(P)^a Q^{ma \deg P}.
\end{aligned} \tag{2.4}$$

Thus the positive square term in the inert-minus-split race is not inserted heuristically; its sign is forced by subtracting the higher prime powers from (2.3).

Lemma 2.1 (Endpoint decomposition). *For $\epsilon \in \{0, 1\}$, set*

$$c_{g,\lambda}^{(0)} = \frac{2r}{r+1}, \quad c_{g,\lambda}^{(1)} = \frac{2}{r+1}, \tag{2.5}$$

and, for $0 \leq \vartheta \leq \pi$, set

$$b_{g,\lambda}(\vartheta) = \frac{4(1-r^{-1})}{|1-r^{-1}e^{-i\vartheta}|} = \frac{4(1-e^{-\lambda/d})}{\sqrt{1+e^{-2\lambda/d}-2e^{-\lambda/d}\cos\vartheta}}. \tag{2.6}$$

There are fixed phase shifts η_j , depending only on the Frobenius angles, such that

$$\mathcal{E}_{g,t,\lambda}(N) = c_{g,\lambda}^{(N \bmod 2)} + \sum_{j=1}^g b_{g,\lambda}(\vartheta_j) \cos(N\vartheta_j + \eta_j) + o_{g,Q,t,\lambda}(1). \tag{2.7}$$

In particular, no prime powers of exponent at least three and no ramified prime contribute to the endpoint limit.

Proof. Since $\alpha_j Q^m = re^{\pm i\vartheta_j}$, the contribution of one conjugate pair to the first term of (2.4), after multiplication by the normalising factor $2(1-r^{-1})r^{-N}$, is

$$4(1-r^{-1}) \operatorname{Re} \left(\frac{e^{iN\vartheta} - r^{-N}}{1-r^{-1}e^{-i\vartheta}} \right). \tag{2.8}$$

The term containing r^{-N} tends to zero. Writing the argument of the fixed denominator into the phase gives exactly the amplitude (2.6).

It remains to treat the prime powers without discarding their endpoint dependence. If $\pi_Q(\ell)$ is the number of monic irreducibles of degree ℓ , the elementary identity $Q^\ell = \sum_{a|\ell} a\pi_Q(a)$ implies

$$\ell\pi_Q(\ell) = Q^\ell + O(\ell Q^{\ell/2}). \tag{2.9}$$

The finite ramified prime polynomials are precisely the d distinct linear factors of $D_{g,t}$. Hence

$$\sum_{\deg P = \ell} (\deg P) \chi_{g,t}(P)^2 = Q^\ell + O(\ell Q^{\ell/2} + d \mathbf{1}_{\ell=1}). \tag{2.10}$$

The main term in $R_2(N)$ is therefore $\sum_{\ell \leq N/2} r^{2\ell}$. If $N = 2L$, its normalised limit is

$$\lim_{L \rightarrow \infty} 2(1 - r^{-1})r^{-2L} \sum_{\ell=1}^L r^{2\ell} = \frac{2r}{r+1};$$

if $N = 2L + 1$, the additional factor r^{-1} gives $2/(r+1)$. For the error in (2.10), the exponent after normalisation is

$$(2M - \tfrac{1}{2})\ell - MN, \quad \ell \leq N/2.$$

Its maximum is at most $-\min(M, 1/4)N + O_{Q,M}(1)$. The polynomial number of summands is harmless, so the complete square-counting error, including the missing ramified linear factors, is $o(1)$.

Finally use $\ell\pi_Q(\ell) \leq Q^\ell$ in absolute value for $R_{\geq 3}(N)$. A term with $a\ell \leq N$ has, after normalisation, base- Q exponent

$$\ell + am\ell - MN. \quad (2.11)$$

If $m \geq 0$, this is at most $\ell + mN - MN \leq -N/6$, because $\ell \leq N/3$. If $m < 0$ and $1 + 3m \geq 0$, it is again at most $-N/6$; if $1 + 3m < 0$, it is at most $-MN + O_M(1)$. Summing the $O(N^2)$ possible pairs (a, ℓ) proves that the normalised $R_{\geq 3}(N)$ is $O_{Q,M}(N^2 Q^{-\min(M, 1/6)N}) = o(1)$. Combining these estimates with (2.8) proves the lemma. $\square$

We now pass from the deterministic endpoint sequence to its limiting law. The precise independence condition needed below is

$$\vartheta_1, \dots, \vartheta_g, \pi \quad \text{linearly independent over } \mathbb{Q}. \quad (\text{LI})$$

Even without (LI), the endpoint empirical law is genuine rather than conventional. Indeed, let H_t be the closure in $\mathbb{T}^g \times \mathbb{Z}/2\mathbb{Z}$ of the cyclic subgroup generated by

$$(e^{i\vartheta_1}, \dots, e^{i\vartheta_g}, 1).$$

The powers of this generator are equidistributed in H_t with respect to Haar probability: this is the character criterion for a compact monothetic group. Push that Haar measure forward by the continuous cosine-and-center function in (2.7). The $o(1)$ remainder has vanishing Cesaro mean square, so the empirical measures converge in W_2 . This defines the actual law $\nu_{g,t,\lambda}$ for every t . Under (LI), the subgroup H_t is the full product, which yields the independent-phase formula below.

Proposition 2.2 (Exact quenched endpoint law). *Assume (LI). The endpoint empirical measures of $\mathcal{E}_{g,t,\lambda}(N)$,*

$$\frac{1}{X} \sum_{1 \leq N \leq X} \delta_{\mathcal{E}_{g,t,\lambda}(N)},$$

converge in W_2 to

$$\nu_{g,t,\lambda} = \frac{1}{2} \sum_{\epsilon=0}^1 \text{Law}_\Phi \left(c_{g,\lambda}^{(\epsilon)} + \sum_{j=1}^g b_{g,\lambda}(\vartheta_j) \cos \Phi_j \right), \quad (2.12)$$

where the Φ_j are independent and uniform on $[0, 2\pi]$. Consequently the natural density of positive endpoints exists and equals

$$\nu_{g,t,\lambda}((0, \infty)) = \frac{1}{2} \sum_{\epsilon=0}^1 \Pr_\Phi \left(c_{g,\lambda}^{(\epsilon)} + \sum_{j=1}^g b_{g,\lambda}(\vartheta_j) \cos \Phi_j > 0 \right). \quad (2.13)$$

Proof. Condition (LI) says equivalently that $1, \vartheta_1/\pi, \dots, \vartheta_g/\pi$ are rationally independent. Kronecker–Weyl applied separately to $N = 2n$ and $N = 2n + 1$ therefore makes the phase vector $N(\vartheta_1, \dots, \vartheta_g)$ equidistributed on the g -torus along each parity class. The fixed shifts η_j do not change Haar measure, and the two parity classes each have natural density $1/2$. Formula (2.7) gives (2.12); because all variables are bounded for fixed g , the convergence also holds in W_2 , not merely weakly. Each nonzero arcsine summand has an absolutely continuous law, so the limiting finite sum has no atom at zero. The portmanteau theorem then gives (2.13). This is the standard discrete Kronecker–Weyl mechanism behind function-field race laws [4, 2], with the parity coordinate retained explicitly. $\square$

At the critical scaling, $r = e^{\lambda/d}$, so both centers in (2.5) tend to 1. If $\vartheta = 2\pi y/d$ with y in a fixed compact subset of $[0, \infty)$, then directly from (2.6)

$$b_{g,\lambda}(2\pi y/d) \longrightarrow \frac{4\lambda}{\sqrt{\lambda^2 + (2\pi y)^2}} \quad (2.14)$$

locally uniformly in y . This is the coefficient appearing in the hard-edge law (1.5).

3 Density-one linear independence over square fields

We next show that (LI) holds for all but $o(1)$ of the parameters in the simultaneous regime. The square-field hypothesis is used at one specific algebraic point: it makes the normalized Frobenius roots roots of a rational polynomial.

For $t \in U_g(\mathbb{F}_Q)$, write

$$P_{g,t}(T) = L(T, \chi_{g,t}) = \prod_{j=1}^{2g} (1 - \alpha_j T) \in \mathbb{Z}[T]$$

and form its reciprocal polynomial

$$R_{g,t}(X) = X^{2g} P_{g,t}(1/X) = \prod_{j=1}^{2g} (X - \alpha_j). \quad (3.1)$$

Suppose $Q = s^2$, with $s \in \mathbb{Z}_{>0}$. The normalized inverse roots $\beta_j = \alpha_j/s$ are then the roots of

$$\tilde{R}_{g,t}(X) := s^{-2g} R_{g,t}(sX) = \prod_{j=1}^{2g} (X - \beta_j) \in \mathbb{Q}[X]. \quad (3.2)$$

Multiplication of every root by the nonzero rational number s^{-1} changes neither the splitting field nor the permutation action of its Galois group. The roots of (3.2) occur in reciprocal pairs $e^{i\vartheta_j}, e^{-i\vartheta_j}$. Write $W_{2g} = (\mathbb{Z}/2\mathbb{Z})^g \rtimes S_g$ for the signed-permutation Weyl group acting on these g reciprocal pairs.

Lemma 3.1 (Maximal Weyl group implies angle– π independence). *Let $g \geq 2$, let Q be a square, and suppose that the splitting-field Galois group of $R_{g,t}$ is the full Weyl group W_{2g} . Then $\vartheta_1, \dots, \vartheta_g, \pi$ are linearly independent over $\mathbb{Q}$.*

Proof. Kowalski's relation-space theorem [15, Proposition 2.4(2), with $m = 1$] says that, under the maximal W_{2g} hypothesis, the rational space of multiplicative relations among the roots β_j is the reciprocal-pair space. In particular, a rational relation vector has equal coordinates on the two members of every reciprocal pair.

Suppose that $\sum_{j=1}^g a_j \vartheta_j + b\pi = 0$ with rational coefficients. Clearing denominators makes a_j, b integral. Exponentiating twice removes the sign coming from $b\pi$ and gives

$$\prod_{j=1}^g (e^{i\vartheta_j})^{2a_j} = 1. \quad (3.3)$$

In the ordered reciprocal pairs $(e^{i\vartheta_j}, e^{-i\vartheta_j})$, the relation vector in (3.3) has coordinates $(2a_j, 0)$. Membership in the reciprocal-pair space forces $2a_j = 0$ for every j . The original additive relation then gives $b = 0$. Notice that this argument uses only the rational relation space; no saturation assertion about an integral relation lattice is required. $\square$

The required maximal-Galois estimate is already available for precisely our one-parameter pencil. We record its corrected form, since omitting the erratum factor would make the genus dependence false.

Proposition 3.2 (Exceptional parameters). *There is an absolute constant $A > 0$ such that, whenever Q has odd characteristic greater than $2g + 1$,*

$$\#\{t \in U_g(\mathbb{F}_Q) : \text{Gal}(R_{g,t}/\mathbb{Q}) \not\cong W_{2g}\} \leq Ag^2 Q^{1-\gamma_g} \log Q, \quad \gamma_g = \frac{1}{4g^2 + 3g + 5}. \quad (3.4)$$

Consequently, for square Q ,

$$\Pr_{t \in U_g(\mathbb{F}_Q)}(t \text{ fails (LI)}) \ll g^2 Q^{-\gamma_g} \log Q \quad (3.5)$$

as soon as $Q > 4g$, with an absolute implied constant.

Proof. Kowalski's large-sieve theorem for $y^2 = f(x)(x - t)$ gives (3.4) with the stated exponent and an absolute constant [14, Theorem 6.2]. Item 3 of the published erratum [13] inserts the factor g^2 in front of $Q^{1-\gamma_g} \log Q$. Our polynomial f_g is squarefree because the characteristic exceeds $2g + 1$, so the theorem applies. Its statement uses reducibility or splitting-field degree below $2^g g!$. The functional equation already embeds the Galois action in the signed permutation group W_{2g} , whose order is $2^g g!$; hence full degree is equivalent to Galois group W_{2g} . Divide by $|U_g(\mathbb{F}_Q)| = Q - 2g$ and invoke lemma 3.1. $\square$

Corollary 3.3 (Simultaneous density-one LI). *Let Q_g be odd square prime powers of characteristic $p_g > 2g + 1$. If*

$$\frac{\log Q_g}{g^2 \log(g + 2)} \longrightarrow \infty, \quad (3.6)$$

then a uniformly chosen $t_g \in U_g(\mathbb{F}_{Q_g})$ satisfies (LI) with probability $1 - o(1)$.

Proof. Taking logarithms in (3.5) gives at most

$$O(\log g + \log \log Q_g) - \frac{\log Q_g}{4g^2 + 3g + 5}. \quad (3.7)$$

Put $x_g = (\log Q_g)/g^2$. Hypothesis (3.6) says $x_g/\log(g + 2) \rightarrow \infty$, while $\log \log Q_g = \log(g^2 x_g) = o(x_g)$. The negative term in (3.7) therefore dominates, and the expression tends to $-\infty$. $\square$

Finally consider the explicit fields in (1.9). By Bertrand's postulate,

$$2g + 1 < p_g < 2(2g + 1).$$

The minimality of n_g gives

$$g^2 \log^2(g + 2) \leq \log Q_g < g^2 \log^2(g + 2) + 2 \log p_g. \quad (3.8)$$

Thus Q_g is an odd square, its characteristic exceeds $2g + 1$, and (3.6) holds. In particular, this completely specified sequence has the density-one linear independence needed to replace the actual endpoint law by (2.12). The much stronger field growth is later used to absorb the heat-smoothing cost; the present large-sieve step itself is only one component of that quantitative ledger.

4 Quantitative transfer from the pencil to Haar measure

This section compares the empirical Frobenius classes in the pencil with Haar measure after applying the probability-measure-valued race functional. There are two quantitative costs: the compactly supported cohomology of a tame character sheaf and the heat trace of $\mathrm{USp}(2g)$. We keep both costs explicit in g .

4.1 The weight-zero sheaf and a direct character estimate

Write $Q = s^2$ and let $\pi_g : \mathcal{C}_g \rightarrow U_g$ be the smooth proper family obtained from (1.1). We use geometric Frobenius throughout: thus $\mathrm{Fr}_{Q,t}^{\mathrm{geom}}$ acts on $R^1(\pi_g)_* \mathbb{Q}_3$ with characteristic polynomial

$$\det(1 - T \mathrm{Fr}_{Q,t}^{\mathrm{geom}}) = \prod_{j=1}^{2g} (1 - \alpha_{j,t} T), \quad \alpha_{j,t} \alpha_{j+g,t} = Q. \quad (4.1)$$

Thus the normalized Frobenius class has eigenvalues $\alpha_{j,t}/s$. This convention is essential in the following half-Tate twist.

The arithmetic quotient of $\pi_1(U_g)$ is procyclic. Since $3 \nmid s$, there is a continuous constant 3-adic character η_s , geometrically trivial, such that

$$\eta_s(\mathrm{Fr}_Q^{\mathrm{arith}}) = s, \quad \eta_s(\mathrm{Fr}_Q^{\mathrm{geom}}) = s^{-1}. \quad (4.2)$$

Indeed, the arithmetic quotient maps continuously to the closed procyclic subgroup generated by s in $\mathbb{Z}_3^\times$. Put

$$\mathcal{F}_g = R^1(\pi_g)_* \mathbb{Q}_3 \otimes \eta_s.$$

The original cup-product pairing has Frobenius multiplier Q ; after the twist its geometric-Frobenius multiplier is $Q \eta_s(\mathrm{Fr}_Q^{\mathrm{geom}})^2 = 1$. The character η_s has weight -1 and is trivial on the geometric fundamental group. At a degree- n closed point it multiplies every geometric-Frobenius eigenvalue by $s^{-n} = Q^{-n/2}$, so the resulting sheaf is pure of weight zero. Consequently $\mathcal{F}_g$ is lisse of rank $2g$, has the same geometric monodromy $\mathrm{Sp}(2g)$, and its full arithmetic image lies in $\mathrm{Sp}(2g)$. Its Frobenius conjugacy classes in the compact form $\mathrm{USp}(2g)$ are exactly the normalized classes $\Theta_{g,t}$. The geometric monodromy and tameness assertions used here are [12, Theorem 10.1.16 and Lemma 10.1.12].

Let $\mu_{g,Q}$ be the central probability measure on $\mathrm{USp}(2g)$ obtained by putting mass $|U_g(\mathbb{F}_Q)|^{-1}$ on each Frobenius conjugacy class and averaging within that class. Let m_g^{Haar} denote Haar probability measure.

Proposition 4.1 (Quantitative character equidistribution). *If $Q > 4g$, then every nontrivial irreducible character χ_ρ of $\mathrm{USp}(2g)$ satisfies*

$$\left| \int \chi_\rho d\mu_{g,Q} \right| \leq \frac{(2g-1)\sqrt{Q}}{Q-2g} \dim \rho \leq \frac{2(2g-1) \dim \rho}{\sqrt{Q}}. \quad (4.3)$$

Proof. Every irreducible representation of the compact group $\mathrm{USp}(2g)$ is the restriction of the corresponding irreducible algebraic representation ρ of $\mathrm{Sp}(2g)_\mathbb{C}$. After choosing a 3-adic model, apply ρ to $\mathcal{F}_g$. The resulting local system $\rho(\mathcal{F}_g)$ is lisse, tame, pure of weight zero, and has rank $\dim \rho$. Since the geometric monodromy is the full group $\mathrm{Sp}(2g)$ and ρ is nontrivial irreducible, it has no geometric invariants.

The base $U_g = \mathbb{A}^1 - \{1, \dots, 2g\}$ has $\chi_c(U_g) = 1 - 2g$. Compactly supported H^0 vanishes because a global section of a lisse sheaf on the connected nonproper curve U_g cannot have proper support unless it is zero. Poincaré duality identifies H_c^2 with the dual of the geometric invariant space, so it also vanishes. The tame Grothendieck–Ogg–Shafarevich formula [18, Part I, Theorem 1 and (2 bis), pp. 133–134] therefore gives the exact identity

$$\dim H_c^1(U_g \otimes \overline{\mathbb{F}}_Q, \rho(\mathcal{F}_g)) = (2g-1) \dim \rho.$$

The Grothendieck–Lefschetz trace formula and Deligne’s weight bound [8, Theorem 3.3.1] now give

$$\left| \sum_{t \in U_g(\mathbb{F}_Q)} \chi_\rho(\Theta_{g,t}) \right| \leq (2g-1) \dim \rho \sqrt{Q}.$$

Division by $|U_g(\mathbb{F}_Q)| = Q - 2g$ proves the first inequality in (4.3); the second follows from $Q > 4g$. $\square$

4.2 Lipschitz control of the quenched law

For $U \in \mathrm{USp}(2g)$, order its positive eigenangles as $0 \leq \vartheta_1 \leq \dots \leq \vartheta_g \leq \pi$, and define

$$T_{g,\lambda}(U) = \frac{1}{2} \sum_{\epsilon=0}^1 \mathrm{Law}_\Phi \left(c_{g,\lambda}^{(\epsilon)} + \sum_{j=1}^g b_{g,\lambda}(\vartheta_j) \cos \Phi_j \right) \in \mathcal{P}_2(\mathbb{R}). \quad (4.4)$$

This is a central map. We equip $\mathrm{USp}(2g)$ with the bi-invariant metric whose value at the identity is

$$\langle X, Y \rangle = -\frac{1}{2} \mathrm{Tr}(XY). \quad (4.5)$$

On the standard maximal torus this is exactly the Euclidean angle metric.

Lemma 4.2 (Metric on the conjugacy quotient). *Let $U, V \in \mathrm{USp}(2g)$, and order their positive eigenangles as $0 \leq \vartheta_1 \leq \dots \leq \vartheta_g \leq \pi$ and $0 \leq \varphi_1 \leq \dots \leq \varphi_g \leq \pi$. For the metric (4.5), let d_G be the Riemannian distance on the group and let d_{conj} be its quotient distance on conjugacy classes, that is, the infimum of d_G between representatives of the two classes. Then*

$$\sum_{j=1}^g |\vartheta_j - \varphi_j|^2 = d_{\mathrm{conj}}([U], [V])^2 \leq d_G(U, V)^2.$$

Proof. For a compact connected Lie group with a bi-invariant metric, every conjugacy class meets a fixed maximal torus, and the induced orbit metric on the conjugacy quotient is the Euclidean torus metric modulo the affine Weyl group. Indeed, a shortest geodesic between the two compact orbits is orthogonal to both by the first-variation formula; bi-invariance then places its tangent in a common maximal torus. Concretely,

$$d_{\text{conj}}([U], [V])^2 = \min_{w \in W_{2g}, k \in (2\pi\mathbb{Z})^g} \|\vartheta - w\varphi + k\|_2^2.$$

For vectors in the closed alcove $0 \leq \vartheta_1 \leq \dots \leq \vartheta_g \leq \pi$, sign changes and 2π -translations cannot improve on pairing two nonnegative angles, and the rearrangement inequality shows that increasing order is the minimizing permutation. Direct matrix multiplication in (4.5) gives the Euclidean angle norm, proving the equality. The inequality follows because the orbit-distance infimum includes the original representatives U, V . $\square$

Lemma 4.3 (Same-phase Wasserstein coupling). *For ordered positive angle vectors ϑ, φ ,*

$$W_2(T_{g,\lambda}(\vartheta), T_{g,\lambda}(\varphi))^2 \leq \frac{1}{2} \sum_{j=1}^g |b_{g,\lambda}(\vartheta_j) - b_{g,\lambda}(\varphi_j)|^2. \quad (4.6)$$

Moreover,

$$\text{Lip}(T_{g,\lambda}) \leq \frac{\|b'_{g,\lambda}\|_\infty}{\sqrt{2}} \leq \frac{\sqrt{2}(2g+1+\lambda)}{\lambda}. \quad (4.7)$$

Proof. Choose the same fair parity variable and the same phases in the two laws. The centers cancel, while independence and $\mathbb{E} \cos^2 \Phi_j = 1/2$ give exactly the right side of (4.6) as the mean squared displacement of this coupling.

Put $d = 2g + 1$, $a = 1 - e^{-\lambda/d}$, and $r = e^{-\lambda/d}$. Then

$$b_{g,\lambda}(\vartheta) = \frac{4a}{\sqrt{a^2 + 4r \sin^2(\vartheta/2)}}.$$

With $u = 2\sqrt{r} \sin(\vartheta/2)/a$, differentiation and $\sup_{u \geq 0} u(1+u^2)^{-3/2} = 2/(3\sqrt{3}) < 1/2$ give

$$|b'_{g,\lambda}(\vartheta)| \leq \frac{2\sqrt{r}}{a} \leq \frac{2(d+\lambda)}{\lambda};$$

the last inequality is $1 - e^{-x} \geq x/(1+x)$. Finally, lemma 4.2 and (4.6) prove (4.7) on the whole group. $\square$

4.3 The transfer estimate

For probability measures on the Polish space $(\mathcal{P}_2(\mathbb{R}), W_2)$, use

$$d_{\text{BL}}(\sigma, \tau) = \sup_{\substack{\|H\|_\infty \leq 1 \\ \text{Lip}(H) \leq 1}} \left| \int H d\sigma - \int H d\tau \right|.$$

Proposition 4.4 (Arithmetic-to-Haar transfer). *There are an absolute constant C and, for each fixed $\lambda > 0$, a constant C_λ such that, whenever $Q > 4g$,*

$$d_{\text{BL}}((T_{g,\lambda})_* \mu_{g,Q}, (T_{g,\lambda})_* m_g^{\text{Haar}}) \leq C_\lambda g^{-2} + Q^{-1/2} \exp(Cg^2 \log(g+2)). \quad (4.8)$$

Consequently the left side tends to zero whenever $\log Q/(g^2 \log(g+2)) \rightarrow \infty$.

Proof. Fix a test function H in the defining supremum and put $h = H \circ T_{g,\lambda}$. Then h is central, $\|h\|_\infty \leq 1$, and by lemma 4.3 it is $L_{g,\lambda}$ -Lipschitz with $L_{g,\lambda} = O_\lambda(g)$. Apply the heat bridge (A.10) from appendix A with

$$\varepsilon_g = \frac{2(2g-1)}{\sqrt{Q}}, \quad D_g = 2g^2 + g, \quad s = g^{-8}.$$

Its smoothing term is

$$2L_{g,\lambda}\sqrt{2D_g s} = O_\lambda(g^{-2}).$$

By (A.5), the character term is at most

$$Q^{-1/2} \exp(Cg^2 \log(g+2));$$

all fixed numerical factors have been absorbed only after the exact heat estimate is applied. The resulting bound is independent of H , so taking the supremum proves (4.8). Under the stated growth condition the logarithm of the second term is at most $-\frac{1}{2} \log Q + Cg^2 \log(g+2)$, which tends to $-\infty$. $\square$

5 The Haar hard edge and the quenched law

Let U_g be Haar distributed in $\mathrm{USp}(2g)$, and write its positive eigenangles as $0 < \vartheta_1 < \dots < \vartheta_g < \pi$. With $d = 2g + 1$, set

$$\Xi_g = \sum_{j=1}^g \delta_{d\vartheta_j/(2\pi)},$$

regarded as a point measure on the *closed* half-line $[0, \infty)$. Keeping the endpoint is important: vague convergence on $(0, \infty)$ would not see a point moving to zero.

5.1 The finite kernel and its closed-edge limit

Weyl integration, followed by the change of variables $y = d\vartheta/(2\pi)$, gives the projection kernel

$$\tilde{K}_g(x, y) = \frac{4}{d} \sum_{k=1}^g \sin \frac{2\pi kx}{d} \sin \frac{2\pi ky}{d}, \quad 0 \leq x, y \leq \frac{d}{2}. \quad (5.1)$$

Indeed, before rescaling the orthonormal system is $(2/\pi)^{1/2} \sin(k\vartheta)$, $1 \leq k \leq g$. Thus no normalizing constant is hidden in (5.1). For every fixed L , the right side is a Riemann sum, uniformly on $[0, L]^2$, and

$$\tilde{K}_g(x, y) \longrightarrow \frac{2}{\pi} \int_0^\pi \sin(tx) \sin(ty) dt = K_{\mathrm{Sp}}(x, y). \quad (5.2)$$

This convergence includes $x = 0$ and $y = 0$. Moreover,

$$0 \leq \tilde{K}_g(y, y) \leq \frac{4g}{2g+1} < 2, \quad K_{\mathrm{Sp}}(y, y) = 1 - \frac{\sin(2\pi y)}{2\pi y} < 2, \quad (5.3)$$

where the last expression has value zero at $y = 0$.

Extend each finite kernel by zero off $[0, d/2]$. Local uniform convergence first gives convergence of the corresponding operators on compactly supported L^2 test functions. Because all the finite operators and the limit are orthogonal projections, hence contractions, density upgrades this to weak-operator convergence on $L^2([0, \infty))$. Local uniform convergence of the diagonals and the common bound in (5.3) also give convergence of every bounded local trace. The limiting kernel is a locally trace-class orthogonal projection, so the determinantal process exists by Soshnikov's Theorem 3 [19]. His Theorem 5 gives convergence of all local count-cylinder laws. The uniform first-moment estimate $\mathbb{E}\Xi_g([0, L]) \leq 2L$ gives tightness of the restrictions to every compact interval; a diagonal compact-exhaustion argument therefore upgrades the cylinder-law convergence to weak convergence for the vague topology:

$$\Xi_g \implies \Pi_{\text{Sp}} \quad \text{vaguely on } [0, \infty). \quad (5.4)$$

In particular,

$$\sup_g \mathbb{E}\Xi_g([0, \eta]) \leq 2\eta, \quad (5.5)$$

which records explicitly that mass cannot be lost through the hard edge. The limit has no point at zero, since its intensity there is zero.

5.2 Critical coefficients and their tails

For a rescaled angle $y \in [0, d/2]$, the coefficient in the finite random-phase law is exactly

$$\begin{aligned} a_{g,\lambda}(y) &= \frac{4(1 - e^{-\lambda/d})}{\sqrt{(1 - e^{-\lambda/d})^2 + 4e^{-\lambda/d} \sin^2(\pi y/d)}} \\ &= \frac{4 \sinh(\lambda/(2d))}{\sqrt{\sinh^2(\lambda/(2d)) + \sin^2(\pi y/d)}}. \end{aligned} \quad (5.6)$$

On every compact subset of the closed half-line,

$$a_{g,\lambda}(y) \longrightarrow a_\lambda(y) = \frac{4\lambda}{\sqrt{\lambda^2 + (2\pi y)^2}} \quad (5.7)$$

uniformly. There is also a uniform global envelope. Since $\sin(\pi y/d) \geq 2y/d$ for $0 \leq y \leq d/2$, and $d \geq 5$,

$$a_{g,\lambda}(y) \leq \min \left\{ 4, \frac{A_\lambda}{y} \right\}, \quad A_\lambda := \lambda e^{\lambda/10}, \quad (5.8)$$

for $y > 0$. The limiting coefficient obeys the same bound (in fact $a_\lambda(y) \leq 2\lambda/(\pi y)$). Combining (5.3) and (5.8), for every $R > 0$,

$$\sup_g \mathbb{E} \sum_{y \in \Xi_g, y > R} a_{g,\lambda}(y)^2 \leq \frac{2A_\lambda^2}{R}, \quad \mathbb{E} \sum_{y \in \Pi_{\text{Sp}}, y > R} a_\lambda(y)^2 \leq \frac{2A_\lambda^2}{R}. \quad (5.9)$$

Here and below a sum over a point measure counts multiplicity. Notice also that

$$\int_0^\infty a_\lambda(y)^2 dy = 4\lambda.$$

Consequently the limiting squared coefficient sum is finite almost surely, and Markov's inequality turns the first bound in (5.9) into the uniform probability estimate $2A_\lambda^2/(\eta R)$ at threshold $\eta > 0$.

We now apply the closed-hard-edge stability result proved in appendix B. Extend $a_{g,\lambda}$ continuously beyond $d/2$ in any way; this does not change its values on Ξ_g . The local limit (5.7) and the tail estimate (5.9) are precisely the two coefficient hypotheses of proposition B.1. Coupling the same parity variable and the same phases gives

$$W_2^2(T_{g,\lambda}(\mathbf{U}_g), \nu_\lambda(\Pi_{\text{Sp}})) \leq \frac{1}{2} \sum_{\epsilon=0}^1 |c_{g,\lambda}^{(\epsilon)} - 1|^2 + \frac{1}{2} \sum_{j \geq 1} |u_{g,j} - u_j|^2 \quad (5.10)$$

on a Skorokhod coupling, where the two coefficient vectors are matched in increasing point order and padded by zeros. The right side tends to zero in probability. This proves

Proposition 5.1 (Haar quenched limit). *For Haar $\mathbf{U}_g \in \text{USp}(2g)$,*

$$T_{g,\lambda}(\mathbf{U}_g) \Longrightarrow \nu_\lambda(\Pi_{\text{Sp}}) \quad \text{in } (\mathcal{P}_2(\mathbb{R}), W_2).$$

The random measure on the right is Borel measurable and almost surely absolutely continuous.

For clarity, the last assertion does not rest only on the formal stability lemma. The limiting process is locally finite and has no point at zero. It is also infinite almost surely. If $N_L = \Pi_{\text{Sp}}([0, L])$, the determinantal count identities give

$$\mathbb{E}N_L = L + O(1), \quad \text{Var } N_L \leq \mathbb{E}N_L.$$

For each fixed integer k , Chebyshev's inequality yields $\Pr(N_L \leq k) \rightarrow 0$. Monotonicity in L proves infinitude. The configuration therefore has a first point $y_1 > 0$. Conditional on the configuration, the series in (1.5) converges almost surely and in L^2 ; its first summand has a nonzero arcsine density and is independent of the convergent remainder. Their convolution is absolutely continuous.

5.3 Two distinct nondegeneracy statements

We first prove that the random *probability measure* is not deterministic. Every conditional law has mean one and conditional variance

$$V_\lambda(\xi) = \frac{1}{2} \sum_{y \in \xi} a_\lambda(y)^2 = \sum_{y \in \xi} f_\lambda(y), \quad f_\lambda(y) = \frac{8\lambda^2}{\lambda^2 + 4\pi^2 y^2}. \quad (5.11)$$

The representation in (5.2) identifies K_{Sp} as the orthogonal projection, under the unitary sine transform, onto frequencies in $[0, \pi]$. Hence

$$\int_0^\infty |K_{\text{Sp}}(x, y)|^2 dy = K_{\text{Sp}}(x, x). \quad (5.12)$$

Since $f_\lambda \in L^1(0, \infty) \cap L^2(0, \infty)$, compactly supported approximation in the standard determinantal covariance identity is legitimate [19, Definition 3 and (1.20)]. Explicitly, using $K_{\text{Sp}}(x, x) < 2$ and $\rho_2(x, y) \leq K_{\text{Sp}}(x, x)K_{\text{Sp}}(y, y) < 4$,

$$\mathbb{E} \left(\sum_{y \in \Pi_{\text{Sp}}, y > R} f_\lambda(y) \right)^2 \leq 2 \int_R^\infty f_\lambda(y)^2 dy + 4 \left(\int_R^\infty f_\lambda(y) dy \right)^2 \longrightarrow 0.$$

Passing to the limit from compact truncations therefore gives

$$\text{Var } V_\lambda(\Pi_{\text{Sp}}) = \frac{1}{2} \int_0^\infty \int_0^\infty (f_\lambda(x) - f_\lambda(y))^2 |K_{\text{Sp}}(x, y)|^2 dx dy. \quad (5.13)$$

All factors in this formula are essential, including $1/2$. The integral is finite because it is at most $2 \int f_\lambda(x)^2 K_{\text{Sp}}(x, x) dx$. It is strictly positive: f_λ is strictly decreasing on $(0, \infty)$, while the real-analytic kernel is nonzero on an open set off the diagonal. We have proved the following.

Proposition 5.2 (Nondegeneracy of the quenched measure). *The random element $\nu_\lambda(\Pi_{\text{Sp}})$ of $\mathcal{P}_2(\mathbb{R})$ is not almost surely constant.*

Indeed, if that conditional law were fixed, then its variance would be fixed, contrary to (5.13). This argument says nothing by itself about the single scalar $\nu_\lambda(\Pi_{\text{Sp}})((0, \infty))$; we prove that assertion separately.

Proposition 5.3 (Nondegeneracy of the half-line mass). *The random variable*

$$F_\lambda(\Pi_{\text{Sp}}) := \nu_\lambda(\Pi_{\text{Sp}})((0, \infty))$$

is not almost surely constant.

Proof. For a fixed nonempty square-summable configuration, write $S_\xi = \sum_{y \in \xi} a_\lambda(y) \cos \Phi_y$. Its law is symmetric and, by splitting off the first summand, absolutely continuous. Moreover

$$\Pr_\Phi(|S_\xi| < 1) > 0. \quad (5.14)$$

To see this, choose a finite head so that the tail variance is small enough that $\Pr(|S_{\text{tail}}| < 1/2) > 0$, and then require all the finitely many head cosines to be sufficiently close to zero. Symmetry and absolute continuity now give

$$F_\lambda(\xi) = \frac{1}{2} \left(1 + \Pr_\Phi(|S_\xi| < 1) \right) > \frac{1}{2}. \quad (5.15)$$

Choose a compact interval $I \subset (0, \infty)$, bounded away from zero but close enough to it that $a_\lambda(y) \geq a_0 > 2$ on I . For every integer N ,

$$\Pr\{\Pi_{\text{Sp}}(I) \geq N\} > 0. \quad (5.16)$$

Indeed, writing $(n)_N = n(n-1) \cdots (n-N+1)$, the factorial-moment formula [19, Definitions 2 and 3, equations (1.5) and (1.20)] is

$$\mathbb{E}[(\Pi_{\text{Sp}}(I))_N] = \int_{I^N} \det(K_{\text{Sp}}(x_i, x_j))_{i,j \leq N} dx_1 \cdots dx_N.$$

For distinct positive x_i , the determinant is the Gram determinant in $L^2(0, \pi)$ of the functions $t \mapsto (2/\pi)^{1/2} \sin(tx_i)$. These functions are linearly independent: differentiate an identity $\sum_i c_i \sin(tx_i) = 0$ at zero and use the Vandermonde matrix in x_i^2 . The determinant, hence the factorial moment, is positive. This proves (5.16).

On the event in (5.16), select the first N points of $\Pi_{\text{Sp}} \cap I$. Their coefficients lie in $[a_0, 4]$, and their marked sum has characteristic function $\prod_{j=1}^N J_0(a_j t)$, where J_0 is the Bessel function of the first kind of order zero. Uniformly for $a_j \in [a_0, 4]$ and $N \geq 3$, Fourier inversion gives

$$\left\| p_{\sum_{j=1}^N a_j \cos \Phi_j} \right\|_{\infty} \leq \frac{C_{\lambda, I}}{\sqrt{N}}. \quad (5.17)$$

Here are the constants and the inversion step. Uniformly for $a \in [a_0, 4]$, the power series at zero supplies $c, \delta > 0$ such that $|J_0(at)| \leq e^{-ct^2}$ for $|t| \leq \delta$. The strict inequality $|J_0(x)| < 1$ for $x \neq 0$ gives

$$r := \sup_{a \in [a_0, 4], \delta \leq |t| \leq T} |J_0(at)| < 1$$

for every fixed $T > \delta$. Finally choose an absolute B and then T large enough that the standard large-argument bound $|J_0(x)| \leq Bx^{-1/2}$ applies for $x \geq a_0 T$ and $q := B/\sqrt{a_0 T} < 1$. On the positive half-line,

$$\begin{aligned} \int_0^{\delta} e^{-cNt^2} dt &\leq \frac{\sqrt{\pi}}{2\sqrt{cN}}, \\ \int_{\delta}^T r^N dt &\leq Tr^N, \\ \int_T^{\infty} \left(\frac{B}{\sqrt{a_0 t}} \right)^N dt &= \frac{T}{N/2 - 1} q^N \quad (N > 2). \end{aligned}$$

Doubling these bounds proves that the characteristic function is in $L^1(\mathbb{R})$, with norm $O_{\lambda, I}(N^{-1/2})$. Fourier inversion therefore gives a continuous density bounded by $(2\pi)^{-1}$ times that norm, proving (5.17).

Convolving with the independent contribution of all other points cannot increase the density norm. Hence, on the positive-probability crowding event,

$$\frac{1}{2} < F_{\lambda}(\Pi_{\text{Sp}}) \leq \frac{1}{2} + \frac{C_{\lambda, I}}{\sqrt{N}}. \quad (5.18)$$

If $F_{\lambda}(\Pi_{\text{Sp}}) = c$ almost surely, then (5.15) gives $c > 1/2$. Choosing N so large that the right side of (5.18) is smaller than c contradicts (5.16). Thus the scalar is nonconstant. $\square$

6 Assembly, the explicit sequence, and limitations

Let μ_{g, Q_g} be the empirical Frobenius measure of the pencil and let m_g^{Haar} be Haar probability measure on $\text{USp}(2g)$. The preceding sections establish three comparisons:

$$\begin{aligned} \Pr(t_g \text{ is non-LI}) &= o(1), \\ d_{\text{BL}}((T_{g, \lambda})_* \mu_{g, Q_g}, (T_{g, \lambda})_* m_g^{\text{Haar}}) &= o(1), \\ T_{g, \lambda}(\mathbf{U}_g) &\implies \nu_{\lambda}(\Pi_{\text{Sp}}) \quad (\mathbf{U}_g \text{ Haar}). \end{aligned}$$

Here the bounded-Lipschitz distance is taken on the Polish space $(\mathcal{P}_2(\mathbb{R}), W_2)$. Replacing the model law by the actual endpoint law changes the expectation of every bounded-Lipschitz test by at most twice the first probability. The triangle inequality therefore proves (1.7).

The limiting conditional law is almost surely absolutely continuous by Proposition B.1. Hence evaluation on $(0, \infty)$ is almost surely a continuity point for weak convergence of measures. The continuous mapping theorem gives (1.8). Since these masses lie in $[0, 1]$, their expectations converge as well. The two nondegeneracy clauses follow from Propositions 5.2 and 5.3.

This proves theorem 1.1 and corollary 1.2. The explicit sequence in example 1.3 is admissible by the calculation already recorded in (3.8).

Scope. The proof gives a conservative sufficient growth condition, not a practical or optimal threshold. More importantly, it does not establish the fixed-field microscopic limit, nor does it prove an analogous theorem for the full $2g + 1$ -parameter hyperelliptic ensemble. The theorem is a quantitative large-field/large-genus bridge for the explicit pencil (1.1).

A A uniform heat estimate on $\mathrm{USp}(2g)$

We prove the compact-group estimate used in section 4, including the metric, Casimir, and Haar normalizations. Throughout this appendix, $G_g = \mathrm{USp}(2g)$, $m_g^{\mathrm{Haar}}(G_g) = 1$, and the metric is (4.5). If $X_1, \dots, X_{D_g}$ is an orthonormal basis of left-invariant vector fields, where $D_g = 2g^2 + g$, set

$$\mathcal{A}_g = -\sum_{j=1}^{D_g} X_j^2, \quad P_s = e^{-s\mathcal{A}_g}. \quad (\text{A.1})$$

Thus the Euclidean model of this generator has coordinate variance $2s$. We write $K_s(x)$ for the heat-kernel density relative to Haar *probability* measure, not relative to unnormalized Riemannian volume; see also [11].

A.1 Root coordinates and the heat trace

On the standard maximal torus, write

$$X(x) = \mathrm{diag}(ix_1, \dots, ix_g, -ix_1, \dots, -ix_g).$$

Direct multiplication gives

$$-\frac{1}{2} \mathrm{Tr}(X(x)X(y)) = \sum_{i=1}^g x_i y_i. \quad (\text{A.2})$$

The type- C_g positive roots are therefore expressed in the Euclidean basis e_i , and their half-sum is $\varrho = (g, g-1, \dots, 1)$. Irreducibles are indexed by partitions $\gamma = (\gamma_1 \geq \dots \geq \gamma_g \geq 0)$. With the normalization (A.1), the Casimir eigenvalue is exactly

$$\kappa_\gamma = \|\gamma + \varrho\|^2 - \|\varrho\|^2 = \sum_{i=1}^g \gamma_i (\gamma_i + 2(g-i+1)). \quad (\text{A.3})$$

For example, at $g = 1$ the defining representation has $\kappa_{(1)} = 3$, agreeing with $-\sum_{j=1}^3 (i\sigma_j)^2 = 3I$ for the orthonormal basis $i\sigma_1, i\sigma_2, i\sigma_3$. This check rules out a hidden factor of two in the heat time.

Let d_γ be the representation dimension. The type- C_g Weyl formula in these coordinates is

$$d_\gamma = \prod_{i=1}^g \frac{\gamma_i + g - i + 1}{g - i + 1} \times \prod_{1 \leq i < j \leq g} \frac{\gamma_i - \gamma_j + j - i}{j - i} \frac{\gamma_i + \gamma_j + 2g - i - j + 2}{2g - i - j + 2}. \quad (\text{A.4})$$

Lemma A.1 (Uniform type- C heat trace). *There is an absolute constant C such that, for $0 < s \leq 1$,*

$$K_{2s}(e) = \sum_{\gamma} d_\gamma^2 e^{-2s\kappa_\gamma} \leq \exp(Cg^2 \log(g+2)) s^{-(2g^2+g+1)/2}. \quad (\text{A.5})$$

The identity is for the heat-kernel density relative to Haar probability.

Proof. Put $R = \gamma_1$. Equation (A.3) gives $\kappa_\gamma \geq R^2$. Formula (A.4) has exactly g^2 factors; every denominator is at least one and every numerator is at most $2(R+g+1)$. Hence

$$d_\gamma \leq [2(R+g+1)]^{g^2}. \quad (\text{A.6})$$

There are at most $(R+1)^g$ partitions with largest part R , and $R+g+1 \leq (g+1)(R+1)$. With $N = 2g^2 + g$, this gives

$$K_{2s}(e) \leq [2(g+1)]^{2g^2} \sum_{R=0}^{\infty} (R+1)^N e^{-2sR^2}. \quad (\text{A.7})$$

To bound the last sum, first use $(R+1)^N \leq 2^N R^N$ for $R \geq 1$. For a nonnegative unimodal function, its sum over positive integers is at most its integral plus twice its maximum. In the present case

$$\int_0^\infty x^N e^{-2sx^2} dx = \frac{1}{2} (2s)^{-(N+1)/2} \Gamma\left(\frac{N+1}{2}\right), \quad \max_{x \geq 0} x^N e^{-2sx^2} = \left(\frac{N}{4es}\right)^{N/2}.$$

Stirling's upper bound for the displayed Gamma factor, together with $s \leq 1$, therefore yields

$$\sum_{R=0}^{\infty} (R+1)^N e^{-2sR^2} \leq \exp(CN \log(N+2)) s^{-(N+1)/2}$$

for $0 < s \leq 1$. Substitution into (A.7), with $N = 2g^2 + g$, proves (A.5). $\square$

A.2 Peter–Weyl smoothing

The following statement isolates exactly how a character estimate is converted into a bound for a nonspectral test function.

Lemma A.2 (Brownian displacement). *Let M be a compact D -dimensional Riemannian manifold with nonnegative Ricci curvature, let $P_s = e^{s\Delta}$ be its heat semigroup, and let $K_s(x, dy)$ be the corresponding transition law. Then*

$$\int_M d(x, y)^2 K_s(x, dy) \leq 2Ds. \quad (\text{A.8})$$

Proof. Put $r(y) = d(x, y)$. Laplacian comparison gives $\Delta r \leq (D - 1)/r$ off x and its cut locus, and the same inequality holds in the barrier, hence distributional, sense. The chain rule away from that exceptional set therefore gives

$$\Delta r^2 = 2r\Delta r + 2|\nabla r|^2 \leq 2D$$

distributionally on all of M . Apply the positive heat semigroup to this inequality and integrate the identity $\partial_u P_u r^2 = P_u \Delta r^2$ from 0 to s . Smooth bounded approximations of r^2 justify the calculation across the cut locus; on a compact manifold no truncation at infinity is needed. Since $r(x) = 0$, the result is (A.8). $\square$

Proposition A.3 (Heat bridge). *Let μ be a conjugation-invariant probability measure on G_g , and suppose that for every nontrivial irreducible representation ρ ,*

$$\left| \int_{G_g} \chi_\rho d\mu \right| \leq \varepsilon d_\rho. \quad (\text{A.9})$$

If h is central, L -Lipschitz, and $\|h\|_\infty \leq 1$, then for $0 < s \leq 1$,

$$\left| \int h d\mu - \int h dm_g^{\text{Haar}} \right| \leq 2L\sqrt{2D_g s} + \varepsilon K_{2s}(e)^{1/2}. \quad (\text{A.10})$$

Proof. Under Haar probability, irreducible characters form an orthonormal basis for the central square-integrable functions. The Fourier and heat expansions are

$$h = \sum_{\rho} \hat{h}(\rho) \chi_\rho, \quad P_s h = \sum_{\rho} e^{-s\kappa_\rho} \hat{h}(\rho) \chi_\rho, \quad \sum_{\rho} |\hat{h}(\rho)|^2 = \|h\|_2^2. \quad (\text{A.11})$$

The trivial character cancels between μ and m_g^{Haar} . Applying (A.9), Cauchy–Schwarz, and $\|h\|_2 \leq \|h\|_\infty$ therefore gives

$$\left| \int P_s h d(\mu - m_g^{\text{Haar}}) \right| \leq \varepsilon \left(\sum_{\rho} d_\rho^2 e^{-2s\kappa_\rho} \right)^{1/2} = \varepsilon K_{2s}(e)^{1/2}. \quad (\text{A.12})$$

The last equality is exact: relative to Haar probability, an irreducible of dimension d_ρ contributes its d_ρ^2 matrix coefficients to the full heat kernel. In particular, the quantity in (A.12) is $K_{2s}(e)$, not $K_s(e)$, and no Riemannian-volume factor occurs.

It remains to control smoothing. A compact group with a bi-invariant metric has nonnegative Ricci curvature: on its Lie algebra, $\text{Ric}(X, X) = \frac{1}{4} \sum_j \|[X, X_j]\|^2 \geq 0$. Thus lemma A.2, translation invariance, and Cauchy–Schwarz imply

$$\|P_s h - h\|_\infty \leq L\sqrt{2D_g s}.$$

Use this once under μ and once under m_g^{Haar} , and insert (A.12). This proves (A.10). $\square$

B Closed-hard-edge stability of marked laws

This appendix records the measurability and coupling statement used in section 5. Its state space is the space $\mathcal{N}([0, \infty))$ of locally finite integer-valued measures on the closed half-line, with the vague topology. The closed endpoint prevents the loss of a critical coefficient when a point approaches zero.

For $u = (u_j)_{j \geq 1} \in \ell^2$ and $c \in \mathbb{R}$, put

$$\mathcal{L}(c, u) = \text{Law}_\Phi \left(c + \sum_{j \geq 1} u_j \cos \Phi_j \right).$$

Orthogonality makes the partial sums Cauchy in L^2 . Moreover, Kolmogorov's maximal inequality gives

$$\Pr \left\{ \sup_{n \geq N} \left| \sum_{j=N}^n u_j \cos \Phi_j \right| > \epsilon \right\} \leq \frac{1}{2\epsilon^2} \sum_{j \geq N} u_j^2.$$

Choosing a subsequence of N 's for which the right side is summable and using Borel–Cantelli proves almost-sure convergence of the series. If the same phases are used for two vectors, then the exact mean-square cost of this displayed coupling is

$$\mathbb{E}_\Phi \left| (c - d) + \sum_{j \geq 1} (u_j - v_j) \cos \Phi_j \right|^2 = |c - d|^2 + \frac{1}{2} \|u - v\|_2^2. \quad (\text{B.1})$$

It follows that

$$W_2^2(\mathcal{L}(c, u), \mathcal{L}(d, v)) \leq |c - d|^2 + \frac{1}{2} \|u - v\|_2^2. \quad (\text{B.2})$$

Thus $\mathcal{L} : \mathbb{R} \times \ell^2 \rightarrow \mathcal{P}_2(\mathbb{R})$ is continuous. We emphasize that (B.1) is the exact cost of the chosen coupling; (B.2) need not be equality for the optimal Wasserstein coupling.

We next make the conditional-law map globally Borel. Enumerate the atoms of $\xi \in \mathcal{N}([0, \infty))$ increasingly, with multiplicity, as $e_1(\xi), e_2(\xi), \dots$, padding a finite configuration by ∞ . These coordinates are Borel; for example, each is recovered from the counting maps $\xi \mapsto \xi([0, r])$ with rational r . Set $a_\lambda(\infty) = 0$ and

$$A_\lambda(\xi) = (a_\lambda(e_j(\xi)))_{j \geq 1}, \quad \mathcal{D}_\lambda = \left\{ \xi : \sum_{j \geq 1} a_\lambda(e_j(\xi))^2 < \infty \right\}.$$

The domain $\mathcal{D}_\lambda$ is Borel, and A_λ is a Borel map from it to the separable Hilbert space ℓ^2 . Therefore

$$\xi \mapsto \begin{cases} \mathcal{L}(1, A_\lambda(\xi)), & \xi \in \mathcal{D}_\lambda, \\ \delta_1, & \xi \notin \mathcal{D}_\lambda \end{cases} \quad (\text{B.3})$$

is a globally Borel $\mathcal{P}_2(\mathbb{R})$ -valued map. On $\mathcal{D}_\lambda$, its mean is one and its centered second moment is $\|A_\lambda(\xi)\|_2^2/2$.

Proposition B.1 (Closed-hard-edge marked-law stability). *Let Ξ_n, Ξ be random simple locally finite point measures on $[0, \infty)$, and suppose $\Xi_n \Rightarrow \Xi$ vaguely and $\Xi(\{0\}) = 0$ almost surely.*

Let a_n, a be continuous on the closed half-line, with $a_n \rightarrow a$ locally uniformly, and assume that, for every $\eta > 0$,

$$\sum_{x \in \Xi_n} a_n(x)^2 < \infty \quad \text{a.s. for every } n, \quad \sum_{x \in \Xi} a(x)^2 < \infty \quad \text{a.s.},$$

and

$$\lim_{R \rightarrow \infty} \sup_n \Pr \left\{ \sum_{x \in \Xi_n, x > R} a_n(x)^2 > \eta \right\} = 0. \quad (\text{B.4})$$

Enumerate increasingly and pad finite coefficient vectors by zero. Then, on a Skorokhod realization,

$$\sum_{j \geq 1} |a_n(e_j(\Xi_n)) - a(e_j(\Xi))|^2 \rightarrow 0 \quad \text{in probability.} \quad (\text{B.5})$$

If $c_n^{(0)}, c_n^{(1)} \rightarrow c$, then the random probability measures

$$\frac{1}{2} \sum_{\epsilon=0}^1 \mathcal{L}(c_n^{(\epsilon)}, A_n(\Xi_n)) \implies \mathcal{L}(c, A(\Xi))$$

in $(\mathcal{P}_2(\mathbb{R}), W_2)$, where A_n, A are the corresponding ordered coefficient vectors. The exact cost of the coupling using one uniform parity variable and shared phases is

$$\frac{1}{2} \sum_{\epsilon=0}^1 |c_n^{(\epsilon)} - c|^2 + \frac{1}{2} \|A_n(\Xi_n) - A(\Xi)\|_2^2. \quad (\text{B.6})$$

Finally, if Ξ has a least atom x_1 almost surely and $a(x_1) \neq 0$, then the limiting conditional law is almost surely absolutely continuous.

Proof. The space of locally finite point measures is Polish, so Skorokhod's theorem applies. Choose a deterministic $R > 0$ for which $\Xi(\{R\}) = 0$ almost surely; almost every R has this property. Because the limiting measure is simple and has no atom at either endpoint of $[0, R]$, vague convergence on the closed half-line implies that the points in $[0, R]$ eventually have the same cardinality and match in increasing order. Local uniform convergence of a_n then makes the finite squared coefficient difference on $[0, R]$ tend to zero almost surely.

After both remaining tails are padded by zeros, their matching cost is at most

$$2 \sum_{x \in \Xi_n, x > R} a_n(x)^2 + 2 \sum_{x \in \Xi, x > R} a(x)^2. \quad (\text{B.7})$$

First let R grow. The first term is uniformly small in probability by (B.4); the second is small in probability by almost-sure square summability. Then let $n \rightarrow \infty$. This proves (B.5). Applying the parity-and-phase coupling gives exactly (B.6), because all cross terms have zero phase expectation. The convergence assertion follows.

For the last statement, split off the least atom. Conditional on Ξ , the variable $a(x_1) \cos \Phi_1$ has an absolutely continuous arcsine law. It is independent of the remainder, which converges almost surely and in L^2 by square summability. Convolution with the arcsine law makes the full conditional law absolutely continuous. $\square$

Declaration of generative-AI assistance

Anthropic Claude and OpenAI Codex were used during preparation for exploratory derivations, source discovery, Lean formalization of elementary sublemmas, and language editing. These systems are not authors. The mathematical claims, source checks, and final text remain the responsibility of the human author; this draft is not a substitute for independent specialist review.

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